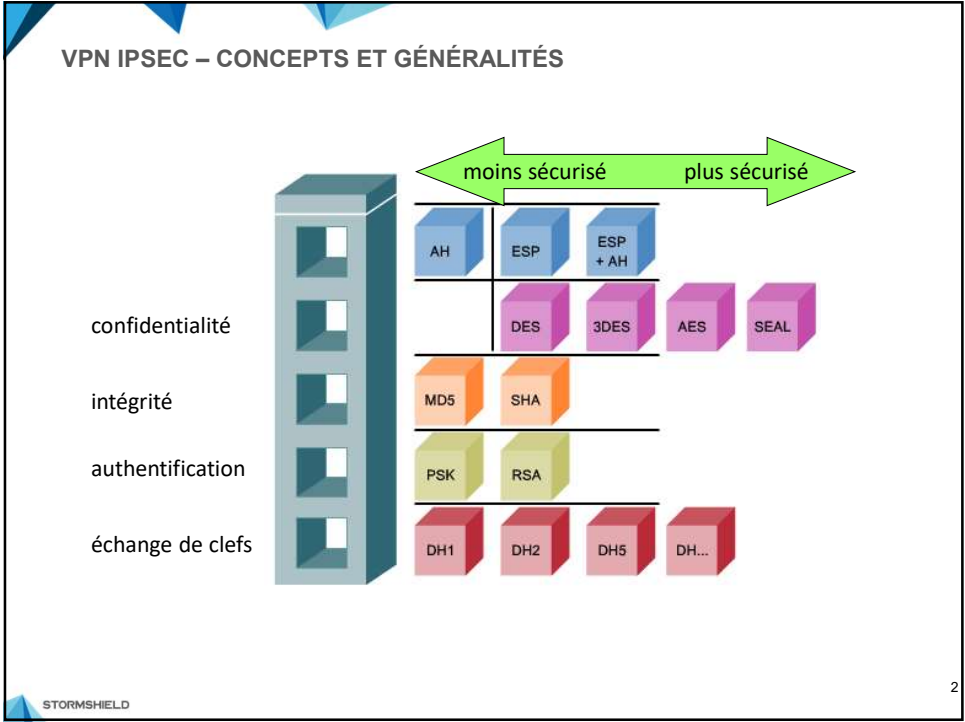
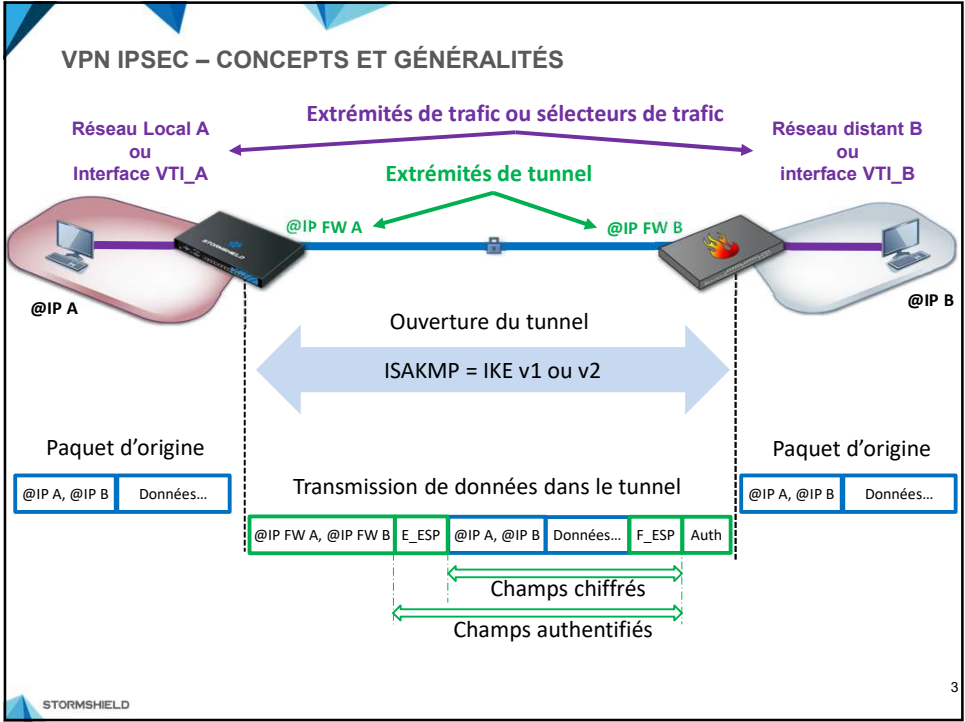




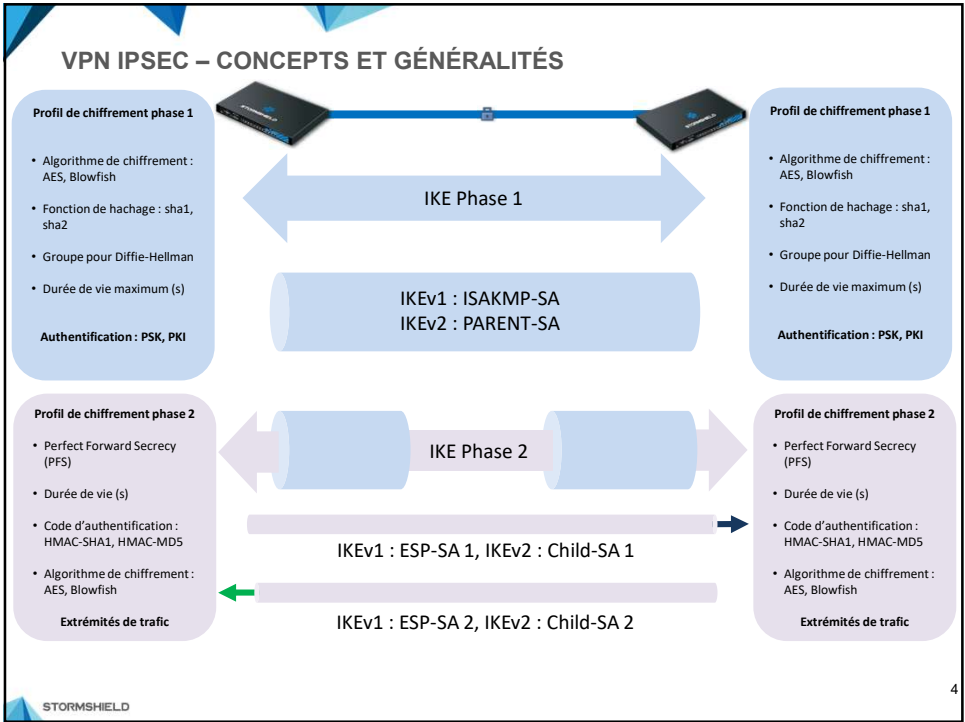
1



2



3



4

- Les différents réseaux privés virtuels
- VPN IPsec – Concepts et généralités
- ➔ **VPN IPsec – Configuration de tunnel site-à-site**
- VPN IPsec – Virtual Tunneling Interface

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Virtual Private Network

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VPN IPSEC - CONFIGURATION D'UN TUNNEL SITE-À-SITE

VPN / IPSEC VPN

ENCRIPTION POLICY - TUNNELS

PEERS

IDENTIFICATION

ENCRIPTION PROFILES

IPsec 01 (01)

Actions

EDIT

RESET

COPY TO

SITE TO SITE (GATEWAY-GATEWAY)

MOBILE - MOBILE USERS

Q Enter a filter

+ Add

X Delete

Up

Down

Cut

Copy

Paste

Show details

(1) IPsec 01

(2) IPsec 02

(3) IPsec 03

(4) IPsec 04

(5) IPsec 05

(6) IPsec 06

(7) IPsec 07

(8) IPsec 08

(9) IPsec 09

(10) IPsec 10

Status

Standard site-to-site tunnel

Remote network

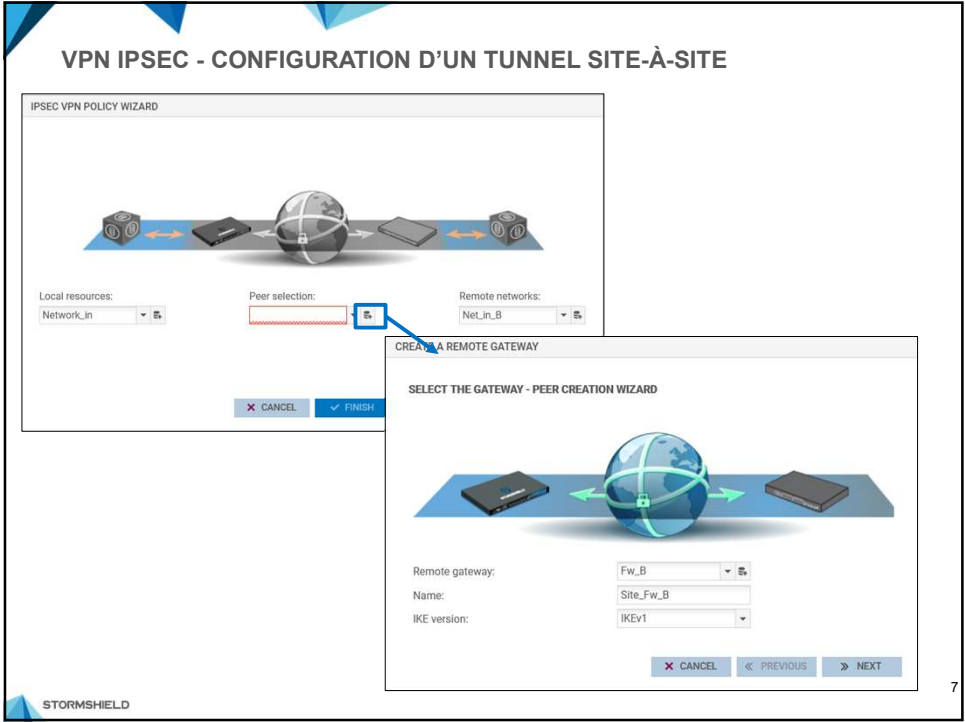
Encryption profile

Separator (rule grouping)

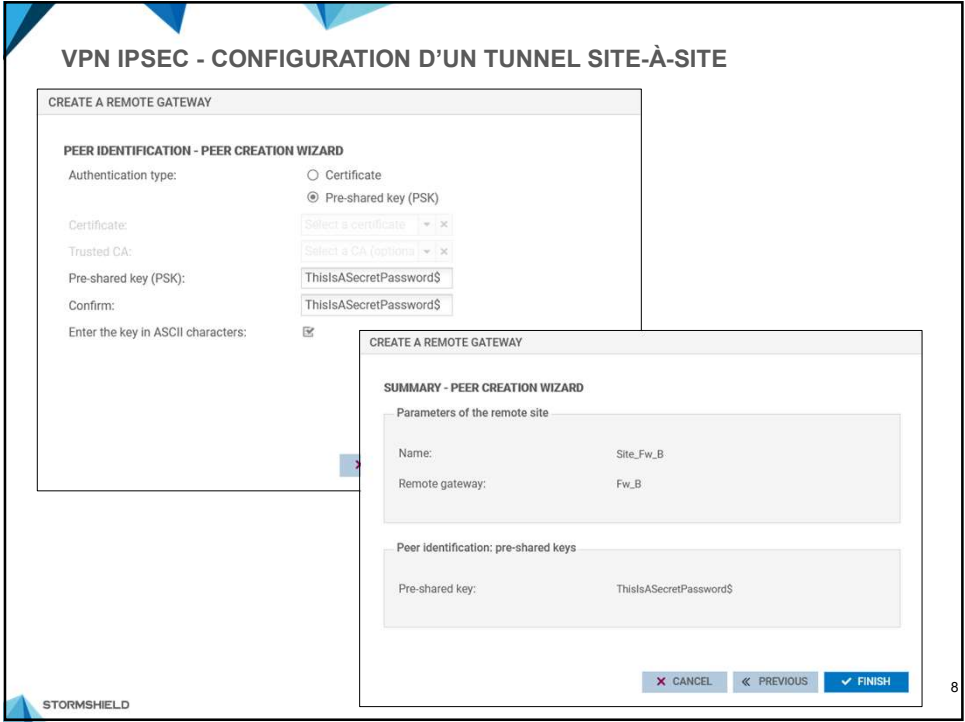
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6



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VPN IPSEC - CONFIGURATION D'UN TUNNEL SITE-À-SITE

IPSEC VPN POLICY WIZARD



Local resources:
Network_In

Peer selection:
Site_Fw_B

Remote networks:
Net_In_B

CANCEL

FINISH

VPN / IPSEC VPN

ENCRIPTION POLICY - TUNNELS

PEERS

IDENTIFICATION

ENCRIPTION PROFILES

IPsec 01 (01)

Actions

1

SITE TO SITE (GATEWAY-GATEWAY)

MOBILE - MOBILE USERS

Enter a filter

+ Add

X Delete

Up

Down

Cut

Copy

Paste

Show details

	Status	Local network	Peer	Remote network	Encryption profile
1	on	Network_In	Site_Fw_B	Net_In_B	StrongEncryption

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VPN IPSEC - CONFIGURATION D'UN TUNNEL SITE-À-SITE

• Les profils de chiffrement phase 1 et phase 2

VPN / IPSEC VPN

ENCRIPTION POLICY - TUNNELS

PEERS

IDENTIFICATION

ENCRIPTION PROFILES

Enter a filter

Remote gateways (1)

Site_Fw_B

General

Comment:

Remote gateway:
Fw_B

Local address:
Any

IKE profile:
StrongEncryption

IKE version:
IKEv1

Identification

Authentication method:
Pre-shared key (PSK)

VPN / IPSEC VPN

ENCRIPTION POLICY - TUNNELS

PEERS

IDENTIFICATION

ENCRIPTION PROFILES

IPsec 01 (01)

Actions

1

SITE TO SITE (GATEWAY-GATEWAY)

MOBILE - MOBILE USERS

Enter a filter

+ Add

X Delete

Up

Down

Cut

Copy

Paste

Show details

	Status	Local network	Peer	Remote network	Encryption profile
1	on	Network_In	Site_Fw_B	Net_In_B	StrongEncryption

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VPN IPSEC - CONFIGURATION D'UN TUNNEL SITE-À-SITE

• Consulter, modifier et créer des profils de chiffrement phase 1 et phase 2

New phase 1 profile

New phase 2 profile

Duplicate

Define the default profile

Delete

Check usage

VPN / IPSEC VPN

ENCIPHERMENT POLICY - TUNNELSPEERSIDENTIFICATIONENCIPHERMENT PROFILES

+ Add

Actions

IKE (4)

DR

GoodEncryption

GoodEncryption

Mobile

IPSec (4)

DR

GoodEncryption

GoodEncryption

Mobile

IKE PROFILE:STRONGENCIPHERMENT

General

Comments:ANSI RGSv2 compliant

Diffie-Hellman:DH14 MODP Group (2048-bits)

Maximum lifetime (in seconds):21600

PROPOSALS

+ Add

Delete

Up

Down

	Algorithm	Strength	Algorithm	Strength
1	aes	256	sha2_256	256
2	aes	128	sha2_256	256

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VPN IPSEC - CONFIGURATION D'UN TUNNEL SITE-À-SITE

• La fonction Keepalive

VPN / IPSEC VPN

ENCIPHERMENT POLICY - TUNNELSPEERSIDENTIFICATIONENCIPHERMENT PROFILES

Separated_CA (01)

Actions

Deactivate

SITE TO SITE (GATEWAY-GATEWAY)

MOBILE - MOBILE USERS

Enter a filter

+ Add

Delete

Up

Down

Cut

Copy

Paste

Show details

Search in logs

Search in r

	Status	Local network	Peer	Remote network	Encryption profile	Keep alive	Comments
1	on	Firewall_out	Columns		StrongEncryption	30	

Columns

Status

Name

Local network

Peer

Remote network

Protocol

Encryption profile

Keep alive

Comments

Keep alive

30

0

30

60

120

300

600

CHECKING THE POLICY

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VPN IPSEC - CONFIGURATION D'UN TUNNEL SITE-À-SITE

- Les règles implicites autorisent le trafic IKE et ESP en provenance du correspondant distant.

SECURITY POLICY / IMPLICIT RULES

IMPLICIT FILTER RULES

Enabled	Name
Enabled	Allow access to the PPTP server
Enabled	Allow mutual access between the members of a firewall cluster (HA)
Enabled	Allow ISAKMP (UDP port 500) and the ESP protocol for IPsec VPN peers.
Enabled	Allow protected interfaces to access the firewall's DNS service (port 53).

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VPN IPSEC - CONFIGURATION D'UN TUNNEL SITE-À-SITE

- Règles de filtrage **explicites** pour autoriser le trafic entre les extrémités de trafic (les réseaux distants)

SECURITY POLICY / FILTER - NAT

(9) Pass all High

Filtering NAT

	Status	Action	Source	Destination	Dest. port	Protocol	Security inspection
1	on	pass	Network_in	NET_IN_B	Any		IPS
2	on	pass	NET_IN_B via IPsec VPN tunnel	Network_in	Any		IPS

EDITING RULE NO 2

General

Source

Destination

Port - Protocol

Inspection

Source

Advanced properties

Source port

Via

IPsec VPN tunnel

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VPN IPSEC - CONFIGURATION D'UN TUNNEL SITE-À-SITE

• Logs de la négociation IKE

LOG / VPN

SEARCH FROM - 07/24/2021 09:52:38 AM - TO - 07/24/2021 10:52:38 AM

Saved at	Message	User	Local network	Destination Name	Source Name	Remote net...	P...	Side	SPI in	SPI out
07/24/2021 10:52:1...	IPSEC SA established		10.0.0.0/24	Remote_peer	Firewall_out	172.29.1.0/24	2	initiator	0xc39cc488	0xc636ed95
07/24/2021 10:52:1...	IKE SA established			Remote_peer	Firewall_out		1	initiator		
07/24/2021 10:52:1...	Charon daemon started						0			
07/24/2021 10:52:1...	Charon configuration reloaded						0			
07/24/2021 10:52:1...	Reloading charon configuration						0			

LOG LINE DETAILS

Configuration

Rule name049ec7261960d1bd4b5854bf1468...

Rule typegateway

Dates

Saved at07/24/2021 10:52:15 AM

Date and time07/24/2021 10:52:15 AM

Time difference between local time ...+0000

Destination

Destination NameRemote_peer

Destination172.18.18.1

Remote network172.29.1.0/24

Remote identifier172.18.18.1

Message

MessageIPSEC SA established

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VPN IPSEC - CONFIGURATION D'UN TUNNEL SITE-À-SITE

• La politique VPN IPSEC

MONITOR / IPSEC VPN TUNNELS

RefreshConfigure the IPsec VPN service

POLICIES

Type	Status	Local traffic endpoint	Local gateway	Local ID	Remote gateway	Peer ID	Remote traffic endpoint
Type : Site-to-site tunnels (2)							
OK	OK	Network_in	Firewall_out	172.18.18.14	Remote_peer	172.18.18.1	remote_net
No tunnels	No tunnels	Network_in	Firewall_out	172.18.18.14	remote_gre	92.61.113.70	remote_net
Type : Exception policies (bypass) (1)							
Bypass		rfc5735_loopback					any

Security Association (SA) IKE

Status:established

Local gateway:Firewall_out

Remote gateway:Remote_peer

Side:responder

Local ID:172.18.18.14

Peer ID:172.18.18.1

Lifetime lapsed:14m

NAT-T:none

Authentication:sha2_256

Encryption:aes/256

PRF:sha256

PFS:14

Security Association (SA) IPsec

Status:installed

Local gateway:Firewall_out

Remote gateway:Remote_peer

Bytes in:

Bytes out:4.43 KB

Lifetime lapsed:14m

Authentication:hmac_sha256

Encryption:aes/256

ESN:Enabled

UDP encapsulation:Disabled

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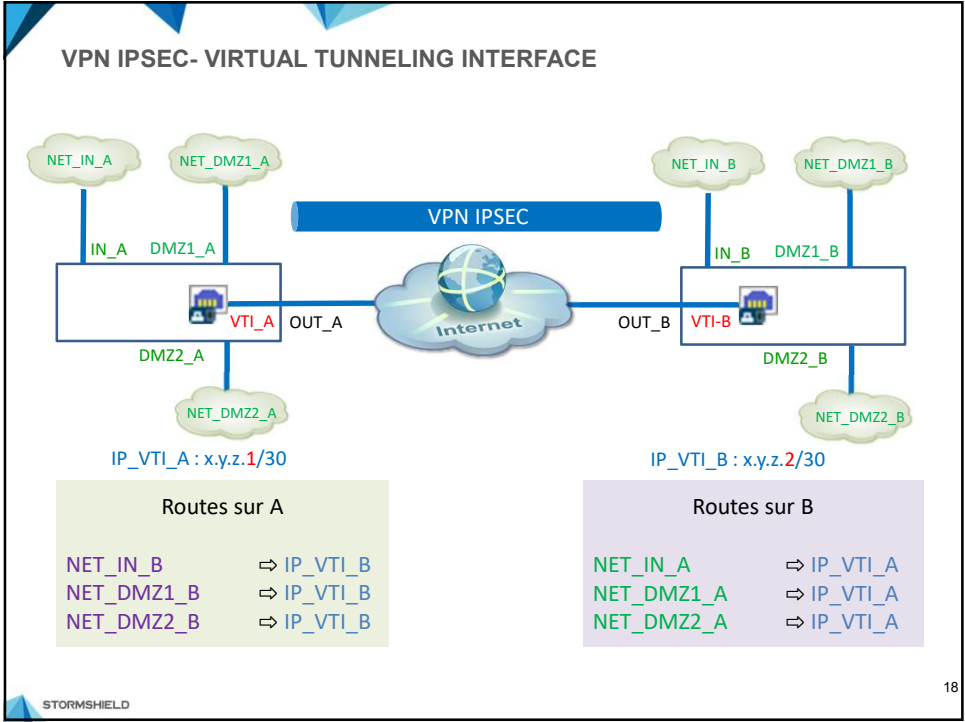
- Les différents réseaux privés virtuels
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- **VPN IPsec – Virtual Tunneling Interface**

→

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VPN IPSEC- VIRTUAL TUNNELING INTERFACE

• Comparatif

Paramètre	VPN par politique	VPN par route (VTI)
Maintient de la SA	Tant que du trafic correspond à la politique	Tant que les vti sont actives
Compatibilité avec le routage dynamique	Non	Oui
Compatibilité multi-éditeur	Oui	Non
Addition d'une nouvelle extrémité de trafic distante à joindre de manière sécurisée	Nécessite la création d'un nouveau tunnel	Nécessite l'ajout d'une route via le tunnel existant
Cas d'usage	- VPN IPsec avec des éditeurs tierce partie - Une seule extrémité de trafic sur le réseau distant	- Répartition de charge avec objet routeur - Tolérance aux pannes par l'usage de routage dynamique - Multiples extrémités de trafic

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VPN IPSEC- VIRTUAL TUNNELING INTERFACE

• Création des interfaces VTI sur chacun des correspondants

Création la VTI sur le correspondant A

NETWORK / VIRTUAL INTERFACES

IPSEC INTERFACES (VTI)

+ Add

X Delete

Check usage

Status	Name	IPv4 address	IPv4 mask
Enabled	VTI_A	172.25.255.1	255.255.255.252

OBJECTS / NETWORK OBJECTS

Firewall_VTI_A

+ Add

X Delete

Check usage

Export

Import

Go

Type	Usage	Name	Value
Type: Hosts (1)			
		Firewall_VTI_A	172.25.255.1 / static

Création de la VTI sur le correspondant B

NETWORK / VIRTUAL INTERFACES

IPSEC INTERFACES (VTI)

+ Add

X Delete

Check usage

Status	Name	IPv4 address	IPv4 mask
Enabled	VTI_B	172.25.255.2	255.255.255.252

OBJECTS / NETWORK OBJECTS

Firewall_VTI_B

+ Add

X Delete

Check usage

Export

Import

Go

Type	Usage	Name	Value
Type: Hosts (1)			
		Firewall_VTI_B	172.25.255.2 / static

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VPN IPSEC- VIRTUAL TUNNELING INTERFACE

Création de l'objet machine qui porte l'adresse IP de l'interface VTI du correspondant distant

Sur A, création de l'objet machine qui porte l'adresse IP de l'interface VTI_B

Sur B, création de l'objet machine qui porte l'adresse IP de l'interface VTI_A

OBJECTS / NETWORK OBJECTS

IP_VTI_B

+ Add X Delete Check usage Export Import Co

Type	Usage	Name	Value
Type : Hosts (1)			
		IP_VTI_B	172.25.255.2 / static

OBJECTS / NETWORK OBJECTS

IP_VTI_A

+ Add X Delete Check usage Export Import Co

Type	Usage	Name	Value
Type : Hosts (1)			
		IP_VTI_A	172.25.255.1 / static

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VPN IPSEC- VIRTUAL TUNNELING INTERFACE

Définition de la politique VPN IPsec basée sur les VTI

Sur le correspondant A

Sur le correspondant B

SITE-TO-SITE (GATEWAY-GATEWAY) ANONYMOUS - MOBILE USERS

Searched text

+ Add X Delete Up Down Cut Copy Paste

Line	Status	Local network	Peer	Remote network	Encryption profile
1	on	Firewall_VTI_A	Site_FW_B	IP_VTI_B	StrongEncryption

SITE-TO-SITE (GATEWAY-GATEWAY) ANONYMOUS - MOBILE USERS

Searched text

+ Add X Delete Up Down Cut Copy Paste

Line	Status	Local network	Peer	Remote network	Encryption profile
1	on	Firewall_VTI_B	Site_FW_A	IP_VTI_A	StrongEncryption

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VPN IPSEC- VIRTUAL TUNNELING INTERFACE

• Définition des routes pour les flux usagers du tunnel

routes statiques

• Sur le correspondant A

STATIC ROUTES

Searching...

+ Add

X Delete

Status	Destination network (host, network or group object)	Interface	Address range	Gateway
on	NET_DMZ1_B	VTI_A	172.16.2.0/24	IP_VTI_B
on	NET_IN_B	VTI_A	192.168.2.0/24	IP_VTI_B

• Sur le correspondant B

STATIC ROUTES

Searching...

+ Add

X Delete

Status	Destination network (host, network or group object)	Interface	Address range	Gateway
on	NET_DMZ1_A	VTI_B	172.16.1.0/24	IP_VTI_A
on	NET_IN_A	VTI_B	192.168.1.0/24	IP_VTI_A

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VPN IPSEC- VIRTUAL TUNNELING INTERFACE

• Autorisation du trafic entre les deux réseaux distants

FILTERING NAT

Searching...

+ New rule

X Delete

↑

↓

↕

↕

Cut

Copy

Paste

Search in logs

Search in monitoring

	Status	Action	Source	Destination	Dest. port	Protocol	Security inspection
1	on	pass	Network_in	NET_IN_B	Any		IPS
2	on	pass	NET_IN_B interface: VTI_A	Network_in	Any		IPS

EDITING RULE NO 2

General

Source

Destination

Port - Protocol

Inspection

SOURCE

GENERAL GEOLOCATION / REPUTATION

General

User:

Source host:

Incoming interface: VTI_A

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RECOMMANDATIONS

- Utiliser des algorithmes robustes pour IKE et IPsec
- Utiliser IKEv2
 - ↓ A défaut utiliser le mode MAIN de IKEv1
- Utiliser des authentifications par certificat
 - ↓ A défaut utiliser des PSK robustes
- Configurer KeepAlive